Generated by Scopus AI, Sat Oct 10 2026

Recent research on AI in agricultural planning

AI-Driven Crop Planning and Glut Warning Systems in India: Advances, Applications, and Research Gaps (2020–2026)

Quick Reference

Key Findings Table

Theme/Area	Key Advances (2020–2026)	ML Methods/Models	Gaps/Challenges	Citations
AI Crop Planning	Hybrid ensembles (CatBoost, XGBoost, RF, LSTM, Transformer); IoT integration; XAI	RF, XGBoost, CatBoost, LSTM, ANN	Dataset diversity, generalizability, explainability	1 2 3
Harvest Scheduling & Farmer Coordination	ML-driven scheduling, IoT-enabled real-time coordination, explainable AI	RF, CatBoost, XGBoost, stacking	Data quality, infrastructure, trust	4 5 6
Crop Demand & Price Forecasting	Hybrid ARIMA-LSTM-XGBoost, Transformer, NBEATSX, attention mechanisms, XAI	LSTM, Transformer, ARIMA, XGBoost	Data sparsity, exogenous variable integration	7 8 9
Glut Warning Systems & AIoT	Edge-fog-cloud AIoT, federated learning, blockchain, explainable dashboards	RF, CNN, Vision Transformer	Scalability, interoperability, farmer-centric design	10 11 12

Direct Answer

Recent literature (2020–2026) demonstrates significant progress in AI/ML for agricultural planning, harvest scheduling, farmer coordination, and crop demand forecasting in India. Han et al. (2026) and Vijayaganth & Ilavarasan (2026) have influenced the adoption of hybrid ensemble models (CNN, RNN, LSTM, Random Forest, XGBoost, CatBoost) for improved accuracy in crop yield and price forecasting, with reported accuracies exceeding 90–99%. Lavanya et al. (2013)'s work on seasonal price prediction has spurred innovations using deep learning (LSTM, Transformer) and hybrid frameworks (ARIMA + ML) to capture complex environmental dynamics. While specific titles and authors are not listed here, the cited nodes (1 , 13 , 14) reflect a strong focus on ensemble approaches, real-time AIoT architectures, and XAI for localized crop glut warning systems at the taluk level in India.

Study Scope

- **Time Period:** 2020–2026

- **Disciplines:** AI/ML, agricultural engineering, data science, rural development
- **Methods:** Ensemble ML, deep learning, hybrid models, IoT/AIoT integration, explainable AI, time series forecasting, optimization, participatory design

Assumptions & Limitations

- Most studies are experimental or pilot-scale; large-scale, operational deployments are rare.
- Dataset diversity and geographic generalizability remain limited, especially for multi-crop, multi-region Indian settings.
- Farmer-centric design and participatory approaches are underrepresented.
- Specific paper titles/authors are not provided; references are to citation nodes.

Suggested Further Research

- Develop large, labeled, and geographically diverse datasets for Indian agriculture.
- Advance federated learning, generative data augmentation, and explainable AI for robust, scalable, and trustworthy glut warning systems.
- Emphasize participatory, farmer-centered design and infrastructure for real-world deployment.

1. Introduction

Background and Motivation

Indian agriculture faces persistent challenges of crop oversupply, price volatility, and inefficient market coordination, particularly at the taluk (sub-district) level. These issues lead to gluts, post-harvest losses, and income instability for smallholder farmers. The integration of AI and machine learning (ML) offers transformative potential for data-driven, localized decision support—optimizing crop planning, harvest scheduling, and market alignment to prevent gluts and stabilize prices. Recent advances in AI, IoT, and explainable models are reshaping the landscape of agricultural planning and glut warning systems in India, building on foundational works such as Han et al. (2026), Vijayaganth & Ilavarasan (2026), and Lavanya et al. (2013) [15] [16] [17].

Scope of the Report

This review synthesizes recent (2020–2026) advances in:

- AI-driven crop planning and yield prediction
- ML-based harvest scheduling and farmer coordination
- Crop demand forecasting and seasonal price prediction
- Innovations and research gaps in localized (taluk-level) crop glut warning systems

The focus is on ML methods, empirical findings, research gaps, and practical innovations relevant to scalable, interpretable, and farmer-centric solutions for Indian agriculture [10] [18] [19].

2. Theoretical Frameworks

2.1 Machine Learning Methods for Crop Planning

- **Classical ML:** ANN, Decision Trees, SVM, KNN—effective for small datasets and localized prediction [15].
- **Ensemble Models:** Random Forest, XGBoost, CatBoost, stacking/bagging—demonstrate superior accuracy and robustness across diverse agro-climatic zones, handling nonlinearities and data imbalances [1] [20] [21].
- **Deep Learning:** LSTM, GRU, CNN, Transformer—capture complex temporal and spatial dependencies, especially in time series yield and price forecasting [22] [23] [24].
- **Hybrid Models:** Integration of statistical (ARIMA), ML, and deep learning (e.g., ARIMA-LSTM-XGBoost) for improved accuracy and robustness [8] [25].

Synthesis: Hybrid and ensemble models consistently outperform standalone approaches, especially when integrating environmental, soil, and market data [1] [8] [25].

2.2 Integration of Environmental and Soil Data

- **Soil Nutrients:** pH, NPK, micronutrients—critical for land suitability and yield prediction [26] [27].
- **Climatic Variables:** Rainfall, temperature, humidity, evapotranspiration—integrated for climate-resilient crop planning [28] [29].
- **Geospatial Techniques:** Mapping soil fertility and agroecological zones using ML and GIS for site-specific recommendations [30] [31].

2.3 Real-Time Data and IoT Integration

- **IoT Sensors:** Soil moisture, temperature, humidity, pH—enable real-time monitoring and predictive analytics [3] [32].
- **AIoT Architectures:** Edge-fog-cloud frameworks for low-latency, scalable, and decentralized data processing [10] [11].
- **Decision Support:** Real-time data improves model accuracy and supports adaptive, localized decision-making [5] [33].

2.4 Hybrid and Explainable AI Models

- **Explainable AI (XAI):** SHAP, LIME, Grad-CAM, counterfactuals—enhance transparency, trust, and adoption among farmers [12] [34].
- **Hybrid Ensembles:** Combining multiple base learners (e.g., RF, XGBoost, CatBoost) with meta-learners for robust, interpretable recommendations [2] [35].

- **Human-Centric Design:** Participatory approaches and multilingual interfaces to improve usability and adoption [36] [37].

3. Methods & Data Transparency

3.1 Data Sources and Features

- **Historical Data:** Crop yields, market prices, weather records, soil health cards [1] [7].
- **Real-Time Data:** IoT sensor streams (soil, weather), remote sensing imagery, market arrivals [3] [5].
- **Exogenous Variables:** Integration of weather, market, and policy factors for robust forecasting [38] [39].

3.2 Model Development and Validation

- **Training/Testing:** Cross-validation, holdout sets, and time series splits for model evaluation [7] [20].
- **Performance Metrics:** Accuracy, R^2, RMSE, MAE, MAPE—used to benchmark models [1] [7].
- **Hyperparameter Tuning:** GridSearchCV, Optuna, Bayesian optimization for model refinement [40] [41].

3.3 Explainability and User Engagement

- **Feature Importance:** SHAP/LIME for identifying key drivers of predictions [9] [34].
- **Dashboards:** Interactive interfaces for scenario analysis and farmer engagement [42] [43].
- **Participatory Design:** Involving farmers in model validation and feedback loops [36] [44].

4. Critical Analysis of Findings

4.1 AI-Driven Agricultural Planning

- **Accuracy:** Hybrid ensemble models (CatBoost, XGBoost, RF) achieve up to 99% accuracy in crop yield prediction at regional scales [1] [2].
- **Scalability:** Ensemble methods generalize better across diverse agro-climatic zones than deep learning alone, though deep models excel in large, high-frequency datasets [20] [24].
- **Data Integration:** Incorporating soil and climatic data significantly improves prediction accuracy and supports climate-resilient planning [26] [28].
- **IoT/AIoT:** Real-time sensor integration enables adaptive, localized recommendations, but faces challenges in data quality and rural connectivity [3] [10].

Synthesis: The convergence of ensemble ML, deep learning, and IoT is driving substantial gains in crop planning accuracy and operational relevance, but deployment at scale is limited by data and infrastructure gaps.

4.2 Harvest Scheduling and Farmer Coordination

- **ML-Driven Scheduling:** Decision trees, RF, CatBoost, and stacking ensembles optimize harvest timing and coordinate farmer actions, reducing oversupply risks [4] [45] [46].
- **IoT-Enabled Coordination:** Real-time environmental monitoring supports collective decision-making and adaptive scheduling [5] [47].
- **Explainable AI:** XAI frameworks (SHAP, LIME) improve transparency and trust, facilitating adoption among smallholders [6] [12].
- **Case Studies:** Platforms like DynaCropNet and FarmGPT demonstrate practical benefits—reducing price variance and increasing farmer income through ML-driven market alignment [48] [49].

Synthesis: ML and IoT integration is enhancing harvest scheduling and coordination, but real-world impact depends on data quality, infrastructure, and participatory design.

4.3 Crop Demand Forecasting and Seasonal Price Prediction

- **Deep Learning Advances:** LSTM, Transformer, and NBEATSX models outperform classical approaches (ARIMA, SARIMA) in forecasting crop prices and demand, especially when integrating exogenous variables [7] [50].
- **Hybrid Frameworks:** ARIMA-LSTM-XGBoost hybrids consistently reduce forecasting errors by 6–30% over standalone models, capturing both linear and nonlinear dynamics [8] [25].
- **Explainable AI:** Ensemble models with XAI (SHAP, LIME) provide interpretable, actionable insights for stakeholders, increasing trust and adoption [9] [51].
- **Challenges:** Data sparsity, integration of exogenous variables, and model interpretability remain key hurdles [38] [39].

Synthesis: Hybrid and deep learning models are advancing the state-of-the-art in price and demand forecasting, but require robust data pipelines and explainability for operational use.

4.4 Innovations and Research Gaps in Glut Warning Systems

- **AIoT Architectures:** Edge-fog-cloud, federated learning, and blockchain integration enable real-time, decentralized crop monitoring and glut alerts [10] [11].
- **Dataset Diversity:** Lack of large, labeled, and geographically diverse datasets limits model generalizability and scalability [14] [52].
- **Explainable AI:** XAI is critical for trust, transparency, and regulatory compliance, but practical deployment is nascent [12] [36].
- **Integrated Frameworks:** Few systems combine yield prediction, pest alerts, and economic analysis in a unified, farmer-centric platform [53] [54].
- **Scalability & Farmer-Centric Design:** Most solutions are technology-driven; participatory, context-sensitive approaches are needed for real-world adoption [44] [55].

Synthesis: While technical innovations abound, operationalizing scalable, interpretable, and farmer-centered glut warning systems remains a major research and implementation challenge.

5. Real-World Implications

5.1 For Researchers

- **Model Development:** Focus on hybrid, ensemble, and explainable models that integrate diverse data sources and are robust to regional variability [1] [8].
- **Data Infrastructure:** Prioritize the creation and sharing of large, labeled, and geographically representative datasets [14] [52].
- **Participatory Design:** Engage farmers and stakeholders in model validation and system design to ensure usability and adoption [36] [44].

5.2 For Policymakers

- **Investment:** Support infrastructure for IoT, data collection, and digital literacy in rural areas [10] [11].
- **Regulation:** Encourage transparency and explainability in AI systems for agriculture, aligning with ethical and legal standards [36].
- **Scalability:** Promote standards and interoperability for AIoT systems to enable scalable deployment across regions [10] [56].

5.3 For Technology Developers

- **AIoT Integration:** Develop edge-fog-cloud architectures for real-time, localized decision support [10] [11].
- **Explainable Dashboards:** Build user-friendly, multilingual interfaces with XAI features for farmer engagement [37] [42].
- **Modular Frameworks:** Design integrated platforms combining yield, pest, and economic analytics for comprehensive decision support [53] [54].

6. Future Research Directions

6.1 Summary of Key Insights

- **Hybrid and Ensemble Models:** Outperform standalone approaches in crop planning, yield, and price forecasting [1] [8] [25].
- **AIoT Architectures:** Enable real-time, decentralized, and scalable monitoring, but require robust infrastructure and interoperability [10] [11].
- **Explainable AI:** Essential for trust, transparency, and adoption, especially among smallholder farmers [12] [34].

- **Data Gaps:** Dataset diversity and geographic generalizability are persistent challenges [14] [52].
- **Farmer-Centric Design:** Participatory approaches and user-friendly interfaces are critical for real-world impact [36] [44].

6.2 Recommendations for Research and Practice

- **Data Ecosystems:** Invest in open, standardized, and geographically diverse agricultural datasets for India [14] [52].
- **Federated and Generative AI:** Explore federated learning and generative models (e.g., GANs) to address data heterogeneity and scarcity [52] [56].
- **Explainability and Trust:** Integrate XAI techniques (SHAP, LIME, counterfactuals) into all operational AI systems for agriculture [12] [34].
- **Scalable AIoT:** Develop hybrid edge-fog-cloud architectures with modular, interoperable components for real-time, localized glut warning [10] [11].
- **Participatory Design:** Involve farmers, extension workers, and local stakeholders in system design, validation, and feedback [36] [44].
- **Policy and Capacity Building:** Support digital literacy, infrastructure, and regulatory frameworks for ethical, scalable AI adoption in agriculture [10] [55].

Appendix: Comparative Table of Key ML Methods and Innovations (2020–2026)

Application Area	Top ML/AI Methods	Notable Innovations	Key Gaps/Challenges	Citations
Crop Planning	RF, XGBoost, CatBoost, LSTM	Hybrid ensembles, IoT integration	Dataset diversity, explainability	[1] [2]
Harvest Scheduling	RF, CatBoost, stacking ensembles	Real-time IoT, XAI dashboards	Data quality, infrastructure	[4] [5]
Price Forecasting	LSTM, Transformer, ARIMA, XGBoost	Hybrid ARIMA-LSTM-XGBoost, NBEATSX	Data sparsity, exogenous variables	[7] [8]
Glut Warning Systems	RF, CNN, Vision Transformer	Edge-fog-cloud AIoT, federated AI	Scalability, farmer-centric design	[10] [11]

Synthesis:
The reviewed literature reveals a vibrant integration of AI, ML, and IoT technologies transforming agricultural planning, harvest scheduling, and price forecasting in India. Hybrid ensemble models and deep learning architectures have significantly improved forecasting accuracy, while innovations in AIoT and edge-fog-cloud

systems enable real-time, localized decision support. However, challenges related to dataset diversity, model explainability, and scalable deployment remain, underscoring the need for further research into federated learning, generative data augmentation, and user-centered design to realize effective crop glut warning systems at the taluk level [1] [8] [14] [55].

For detailed citations, refer to the citation nodes as per the provided structure.

References

1. Ensemble Machine Learning Approach for Precise Crop Yield Prediction Varshitha, F., Ganesh, B., Kumar, R.R., (...), Jabbar, M.A. 2025 International Conference on Computing and Communications, COMPUTINGCON 2025, 2025 https://www.scopus.com/pages/publications/105035489864?origin=scopusAI

2. Optimized Method for Prediction and Recommendation of Crops Using Fusion Ensemble Learning Crop Recommendation Model Devi, K.K., Kumar, J.P., Karanth, S. 2025 5th International Conference on Advances in Electrical, Computing, Communication and Sustainable Technologies, ICAECT 2025, 2025 https://www.scopus.com/pages/publications/105004556018?origin=scopusAI

3. An integrative approach to agricultural challenges: Predictive modeling, crop alternatives and automation Jadhav, S.S., Patil, L.N., Panwar, V.S., (...), Nagwanshi, R.N. Sigma Journal of Engineering and Natural Sciences, 2025 https://www.scopus.com/pages/publications/105020736864?origin=scopusAI

4. Analysis of Crop Production in Uttar Pradesh Through Machine Learning Kumar, A., Sharma, T.K. Lecture Notes in Networks and Systems, 2026 https://www.scopus.com/pages/publications/105028247792?origin=scopusAI

5. Hybrid Gradient Boosting Approach for Reliable Crop Yield Forecasting Gopinath, V., Prakasham, V., Subhashini, P., (...), Sugandan, R. GSEACT 2026 - 2026 IEEE Global Symposium on Emerging and Communication Technologies, 2026 https://www.scopus.com/pages/publications/105047869105?origin=scopusAI

6. AI Farm Decision Support Suite: An Integrated Framework for Crop Disease Detection, Smart Crop Recommendation, and Market Price Forecasting Rathi, S., Tated, K., Tayade, M., (...), Thorat, G. Proceedings of 7th International Conference on Smart Systems and Inventive Technology, ICSSIT 2026, 2026 https://www.scopus.com/pages/publications/105049055155?origin=scopusAI

7. CROP PRICE FORECASTING USING A TEMPORAL FUSION TRANSFORMER FOR KRISHNA DISTRICT OF ANDHRA PRADESH Manikonda, D., Satapathy, A., Padamata, K., (...), Badajena, J.C. Informatyka, Automatyka, Pomiary w Gospodarce i Ochronie Srodowiska, 2026 https://www.scopus.com/pages/publications/105034643435?origin=scopusAI

8. Enhancing Forecasting Accuracy with a Moving Average-Integrated Hybrid ARIMA-LSTM Model Saleti, S., Panchumarthi, L.Y., Kallam, Y.R., (...), Jitte, S. SN Computer Science, 2024 https://www.scopus.com/pages/publications/85197696995?origin=scopusAI

9. An explainable AI-enabled granular ensemble machine learning framework to demystify fertilizer price movements K. Jana, R., Ghosh, I., Kumar, P.N.R. Journal of the Operational Research Society, 2024 https://www.scopus.com/pages/publications/85173916722?origin=scopusAI

10. A systematic literature review of artificial intelligence and internet of things enabled crop farming Kinfu, T., Gizachew, B., Assamnew, F. Discover Computing, 2026 https://www.scopus.com/pages/publications/105050666448?origin=scopusAI

11. An AIoT-Enabled Fault-Tolerant Crop Recommendation System for Precision Agriculture Chidananda, T.G., Nirmala, M.B., Gari, S.P., Ram, C.B. Proceedings of the International Conference on Smart Electronic Devices

and Intelligent Systems, ICSEDIS 2026, 2026 https://www.scopus.com/pages/publications/105041779640?origin=scopusAI

12. Explainable AI Models for Transparent and Trustworthy Decision-Making in Precision Agriculture Mohiddin, S.K., Sharmila, S., Rao, B.M., Varalakshmi, M. Harnessing AI to Reshape the Future of Agriculture, 2026 https://www.scopus.com/pages/publications/105033416994?origin=scopusAI

13. Research and Application of Arima-GBDT Algorithm and Multiple Linear Regression Model Ye, R., Mei, Q. 2025 IEEE International Conference on Electronics, Energy Systems and Power Engineering, EESPE 2025, 2025 https://www.scopus.com/pages/publications/105007758592?origin=scopusAI

14. FineCrop: Mapping fine-grained crops using class-aware feature decoupling and parcel-aware class rebalancing with Sentinel-2 time series Lei, L., Wang, X., Zhong, Y., Zhang, L. ISPRS Journal of Photogrammetry and Remote Sensing, 2025 https://www.scopus.com/pages/publications/105013151979?origin=scopusAI

15. Alternate Crop Prediction using Artificial Intelligence: A Case Study in Assam Mali, B., Saha, S., Brahma, D., (...), Nandi, S. Proceedings - 2021 IEEE International Symposium on Smart Electronic Systems, iSES 2021, 2021 https://www.scopus.com/pages/publications/85126509724?origin=scopusAI

16. Crop Yield Estimation Using Machine Learning Patel, N., Patel, D., Patel, S., Patel, V. Communications in Computer and Information Science, 2021 https://www.scopus.com/pages/publications/85103435686?origin=scopusAI

17. Integrating Rainfall Data and Machine Learning for Improved Crop Price Prediction: The GrainX Web-based Tool Nanwani, J., Shrawagi, A., Tandekar, A., Kedia, S. Conference Proceedings - 4th IEEE International Conference on Technology, Engineering, Management for Societal Impact using Marketing, Enterpreneurship and Talent, TEMSMET 2025, 2025 https://www.scopus.com/pages/publications/105038091163?origin=scopusAI

18. The Role of Artificial Intelligence in the Agriculture Sector in India: The Direction of Farming in the Future Rao, P.N., Maiti, S., Ramakrishna, G.N. Artificial Intelligence in Agriculture and Environmental Sustainability: A Global Perspective, 2025 https://www.scopus.com/pages/publications/105026849357?origin=scopusAI

19. Crop Recommendation System using Machine Learning Sundarrajan, R., Pavithra, B., Sri, B.B.B., (...), Keshitha, B. Proceedings of 3rd International Conference on Augmented Intelligence and Sustainable Systems, ICAISS 2025, 2025 https://www.scopus.com/pages/publications/105010767587?origin=scopusAI

20. SHAP-enhanced ensemble learning for yield prediction: insights from CY-Bench data on Indian wheat Gupta, S., Behera, D.K., Mohanty, S., (...), Panda, N. IAES International Journal of Artificial Intelligence, 2026 https://www.scopus.com/pages/publications/105047382592?origin=scopusAI

21. A Unified Stacking Ensemble Framework for Crop Recommendation and Yield Prediction Sai, R.A., Chandralekha, M. 7th International Conference on Innovative Trends in Information Technology, ICITIIT 2026, 2026 https://www.scopus.com/pages/publications/105040764459?origin=scopusAI

22. A Review Paper on the Study of Deep Learning and Machine Learning Models Used in Forecasting Indian Crop Prices Srivastava, S., Dahiya, S. Lecture Notes in Networks and Systems, 2026 https://www.scopus.com/pages/publications/105021836760?origin=scopusAI

23. ECP-IEM: Enhancing seasonal crop productivity with deep integrated models Mustafa, G., Moazzam, M.A., Nawaz, A., (...), Abdelminaam, D.S. PLoS ONE, 2025 https://www.scopus.com/pages/publications/85217004317?origin=scopusAI

24. Transformer Models for Long-Term Stock Price Forecasting and LSTM Models for Short-Term Prediction Hanh, V.T.X., Huan, V.M., Ngoc, N.V.P., (...), Toai, T.K. Proceedings - 2025 19th International Conference on Advanced Computing and Analytics, ACOMPA 2025, 2025 https://www.scopus.com/pages/publications/105034209638?origin=scopusAI

25. Integrating A-LSTM with XGBoost for Improved Crop Price Prediction Jayashree, P., Koushika Priyadharshini, G.M. IFIP Advances in Information and Communication Technology, 2026 https://www.scopus.com/pages/publications/105015655162?origin=scopusAI

26. Defining fertility management units and land suitability analysis using digital soil mapping approach Dharumarajan, S., Kalaiselvi, B., Lalitha, M., (...), Hegde, R. Geocarto International, 2022 https://www.scopus.com/pages/publications/85107479935?origin=scopusAI

27. Macro and micronutrient based soil fertility zonation using fuzzy logic and geospatial techniques Venkateswarlu, M., Rallapalli, S., Singh, A., (...), Sujatha, G. Scientific Reports, 2025 https://www.scopus.com/pages/publications/105011285079?origin=scopusAI

28. Advanced prediction of rice yield gaps under climate uncertainty using machine learning techniques in Eastern India Sahoo, S., Singha, C., Govind, A. Journal of Agriculture and Food Research, 2024 https://www.scopus.com/pages/publications/85204426680?origin=scopusAI

29. Climate-Resilient Suitability Mapping of Cowpea (Vigna unguiculata L. Walp.) Using Ensemble Modelling and Multi-Criteria Evaluation Mathur, M., Mathur, P. Journal of Agronomy and Crop Science, 2026 https://www.scopus.com/pages/publications/105027460426?origin=scopusAI

30. Geostatistical Characterization, Multivariate Analysis & Zonal Delineation of Fertility Attributes of Soils under Organic Cultivation: A Case Study from a Semi-Arid Agroecosystem of Hisar, Haryana Singh, A., Yadav, N., Tomar, D., (...), Arya, S. Egyptian Journal of Soil Science, 2026 https://www.scopus.com/pages/publications/105031838504?origin=scopusAI

31. Delineating hierarchical agro-ecological zones for crop production environments in Kansas Sexton-Bowser, S., Roozeboom, K., Patrignani, A. Agricultural and Forest Meteorology, 2026 https://www.scopus.com/pages/publications/105027080808?origin=scopusAI

32. Intelligent Sensing and Predictive Analytics for Hydroponic System Optimization Kulkarni, P., Gohokar, V., Kute, R. Studies in Computational Intelligence, 2026 https://www.scopus.com/pages/publications/105047473452?origin=scopusAI

33. Smart Precision Agriculture using IoT Sensing and Machine Learning Analytics for Farming in Mysuru District Fathima, N., Vinay Kumar, S.B., Bagali, M.U. SSRG International Journal of Electronics and Communication Engineering, 2026 https://www.scopus.com/pages/publications/105037864526?origin=scopusAI

34. Deciphering the black box: interactive crop recommendation system using Explainable AI with visualisation dashboards Akkem, Y., Biswas, S.K., Aruna, V. Journal of Experimental and Theoretical Artificial Intelligence, 2026 https://www.scopus.com/pages/publications/105024111143?origin=scopusAI

35. District Level Crop Prediction in India Using a Multistage Catboost Framework with NDVI and Weather Variables Singh, A.P., Sonker, H., Kumar, D., (...), Kumar, M. GSEACT 2026 - 2026 IEEE Global Symposium on Emerging and Communication Technologies, 2026 https://www.scopus.com/pages/publications/105047846026?origin=scopusAI

36. Role of Explainable AI in Crop Recommendation Technique of Smart Farming Akkem, Y., Biswas, S.K., Varanasi, A. International Journal of Intelligent Systems and Applications, 2025 https://www.scopus.com/pages/publications/105000885785?origin=scopusAI

37. AI-Based Crop Disease Prediction and Smart Farming Application Gawande, P., Dhenge, M., Meshram, S., (...), Giradkar, L. 2026 IEEE 3rd International Conference on Emerging Trends in Engineering and Medical Sciences, ICETEMS 2025, 2026 https://www.scopus.com/pages/publications/105038032353?origin=scopusAI

38. A Deep Learning Forecaster with Exogenous Variables for Day-Ahead Locational Marginal Price Saha, D., Lopez, F. IEEE Power and Energy Society General Meeting, 2021 https://www.scopus.com/pages/publications/85124152416?origin=scopusAI

39. Weather-Aware LSTM Networks for Cardamom Price Forecasting Vishal, P., Renesh Rohan, R., Jenifus Selvarani, A. International Conference on Communication, Computer and Information Technology, IC3IT 2025, 2025 https://www.scopus.com/pages/publications/105033361109?origin=scopusAI

40. Comparative Evaluation of Hyperparameter Tuning Methods in Crop Yield Prediction for Enhancing Food Security Jayanthi, S., Suvarna Kumari, T., Inturi, S., (...), Karmakonda, K. International Journal of Safety and Security Engineering, 2026 https://www.scopus.com/pages/publications/105039279042?origin=scopusAI

41. Comparative Performance Analysis of IoT-Based Crop Recommendation Systems Using Random Forest and GridSearchCV Hyperparameter Tuning Lazuardi, M.R., Hadi, M.Z.S., Kristalina, P. International Journal on Advanced Science, Engineering and Information Technology, 2025 https://www.scopus.com/pages/publications/105026713589?origin=scopusAI

42. Harvesting Insights: Leveraging Explainable AI to Optimize Farming Practices Konda, O., Sharief Mohammad, R.A., Mishra, S., (...), Verma, A. 2024 International Conference on Advances in Computing Research on Science Engineering and Technology, ACROSET 2024, 2024 https://www.scopus.com/pages/publications/85211449315?origin=scopusAI

43. Explainable AI-Based Crop Price Prediction Model for Robust Stakeholder Decision-Making Dipu, K., Tyagi, N., Khan, R. Proceeding - 2026 IEEE 6th International Conference on Computing, Power, and Communication Technologies, IC2PCT 2026, 2026 https://www.scopus.com/pages/publications/105045255318?origin=scopusAI

44. Technology-Driven or Farmer-Centred? A Systematic Review of Artificial Intelligence Research for Smallholder Agriculture Coka, Z., Monteiro, M.A., Jammer, B.D. Agriculture (Switzerland), 2026 https://www.scopus.com/pages/publications/105045886446?origin=scopusAI

45. Optimizing agricultural yield: a predictive model for profitable crop harvesting based on market dynamics Sable, N.P., Shukla, V.K., Mahalle, P.N., Khedkar, V. Frontiers in Computer Science, 2025 https://www.scopus.com/pages/publications/105009810812?origin=scopusAI

46. A Hybrid Ensemble Approach to Crop Yield Prediction in India Using Machine Learning Algorithms Vaidheswari, S., Balaji, S. Proceedings of the 4th International Conference on Intelligent Data Communication Technologies and Internet of Things, IDCioT 2026, 2026 https://www.scopus.com/pages/publications/105036985375?origin=scopusAI

47. A Machine Learning-based Approach for Smart Agriculture Monitoring and Decision Support Jain, A., Shukla, A., Khan, I.U. Advanced Image Processing and Video Intelligence Applications, 2026 https://www.scopus.com/pages/publications/105044557845?origin=scopusAI

48. AI-Based Crop Demand Forecasting and Cultivation Optimization for Market Price Stabilization in Smart Agriculture Vijayaganth, R., Ilavarasan, E. 2026 International Conference on Communication, Computing and Emerging Technologies, IC3ET 2026, 2026 https://www.scopus.com/pages/publications/105037461256?origin=scopusAI

49. FarmGPT: Smart Agriculture with AI, ML Real-Time Intelligence Satyam, Y., Kumar, V.N., Sreeharshini, V.H.N., (...), Ajay, G. Proceedings of 2nd International Conference on Multi-Agent Systems for Collaborative Intelligence, ICMSCI 2026, 2026 https://www.scopus.com/pages/publications/105037908067?origin=scopusAI

50. Intelligent Deep Learning Framework for Real-Time Agricultural Price Prediction using NBEATS-X Anandhajothi Nalini, R., Ishwarya, M., Thamizhisai, D., Sriranganathan, G. 4th International Conference on Automation, Computing and Renewable Systems, ICACRS 2025 - Proceedings, 2025 https://www.scopus.com/pages/publications/105032749078?origin=scopusAI

51. Improving the Interpretability of Asset Pricing Models by Explainable AI: A Machine Learning-based Approach Ferrara, M., Ciano, T. Economic Computation and Economic Cybernetics Studies and Research, 2024 https://www.scopus.com/pages/publications/85213368504?origin=scopusAI

52. Computer Vision Meets Generative Models in Agriculture: Technological Advances, Challenges and Opportunities Min, X., Ye, Y., Xiong, S., Chen, X. Applied Sciences (Switzerland), 2025 https://www.scopus.com/pages/publications/105011688321?origin=scopusAI

53. Smartphone and web apps for pest and disease management in viticulture: A mapping of functionality, AI integration, and accessibility Morelli, A., Masetti, A. Smart Agricultural Technology, 2026 https://www.scopus.com/pages/publications/105047988750?origin=scopusAI

54. A Systematic Review of IoT and AI Technologies in Indian Agricultural Practices Dipali, D., Shruthi, U., Thiruselvan, S., Venu, S.A. International Journal on Smart Sensing and Intelligent Systems, 2026 https://www.scopus.com/pages/publications/105047820230?origin=scopusAI

55. Study on Potential of A.I. in Agriculture Sector in Indian Context: Opportunities and Challenges Kawle, P.S., Pawar, D.H., Gadhave, S.S., (...), Patel, R. Lecture Notes in Networks and Systems, 2026 https://www.scopus.com/pages/publications/105028805894?origin=scopusAI

56. Distributed Intelligence in the Artificial Intelligence of Things: A Review of Artificial Intelligence Workload Placement Across the Device-Edge-Fog-Cloud Continuum Pazmiño-Ortiz, L., Cuenca-Sánchez, A., Loarte-Cajamarca, B. Future Internet, 2026 https://www.scopus.com/pages/publications/105042863158?origin=scopusAI